

MUSCLE GUARD

RESEARCH LIBRARY · COMPREHENSIVE EDITION

The research, in full.

14 plain-English articles on GLP-1 medications,
muscle loss, protein, and resistance training.

With sources you can verify yourself.

Compiled by the Muscle Guard research team.

Last updated 6 June 2026 · Free to share.

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Compiled and last reviewed.

Compiled 6 June 2026 by the Muscle Guard research team. We update this PDF when the underlying science updates. Email research@muscleguardglp.com to flag corrections.

A note on what is — and is not — directly cited.

The un-intervened lean-mass-loss share figure of 25-40 percent is directly cited from published DEXA substudies. The directional claim that adequate protein and resistance training *substantially reduce* that share is supported by adjacent (non-GLP-1) intervention trials — a GLP-1-specific intervention trial reporting an exact post-intervention figure is pending. Where this distinction matters, this PDF is explicit about it.

Boundary.

Muscle Guard is a self-tracking companion and coach. Not a medical device. Not medical advice. Not affiliated with any prescriber, telehealth provider, or pharmaceutical manufacturer. Always consult your healthcare provider for personal decisions.

BRANCH 1 · WHAT GLP-1s DO TO YOUR BODY

Do you lose muscle on a GLP-1? What the research shows.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 12-minute read

Without specific intervention, between roughly a quarter and forty-five percent of weight lost on a GLP-1 receptor agonist can be lean tissue rather than fat — depending on the drug, the dose, and the trial.^{[1][5][4]} With adequate protein intake (1.2 g/kg/day or above) and a resistance-training stimulus, that share drops substantially.^{[8][9]} This article walks through what the published trial substudies actually report, why the figure varies, and what changes it.

The headline number, qualified.

The STEP-1 trial of semaglutide 2.4 mg vs placebo for 68 weeks produced a mean body-weight reduction of approximately 15 percent. The DEXA substudy reported lean-body-mass reduction in the region of 13 percent — yielding a lean-mass fraction of total weight loss in the region of 40 to 45 percent.^{[5][2]}

The SURMOUNT-1 trial of tirzepatide 15 mg vs placebo for 72 weeks produced ~21 percent body-weight reduction. The DEXA substudy reported a 33.9 percent reduction in fat mass and a 10.9 percent reduction in lean mass — i.e. lean mass accounted for roughly a quarter of total weight loss.^{[4][6]}

The Neeland et al. (2024) systematic review consolidated body-composition data across the GLP-1 receptor agonist class and concluded that lean-mass-loss share varies meaningfully with the molecule, dose, weight-loss magnitude, baseline body composition, and any concurrent training/protein intervention.^[1]

Why lean mass loss happens.

Two mechanisms drive lean-mass loss in this context:

Calorie and protein deficit. GLP-1 receptor agonists suppress appetite so effectively that most users sustain a meaningful daily energy deficit. Without enough dietary protein, muscle protein synthesis cannot keep pace with breakdown.

Reduced training stimulus. Many users feel fatigued or queasy in titration weeks and train less. The 'use it or lose it' principle is unforgiving.^[9]

What changes the figure.

Two interventions, both tested in adjacent (non-GLP-1) energy-deficit trials and supported by GLP-1-era observational data, substantially reduce the lean-mass-loss share:

Protein at or above 1.2 g/kg/day. The Nunes et al. (2022) meta-analysis pooled trials in healthy adults and found that intakes above 1.05 g/kg supported up to 1.21 kg additional fat-free mass retention vs lower intakes.^[8] Longland et al. (2016) compared 1.2 vs 2.4 g/kg/day during a 40 percent deficit with concurrent training; the higher-protein arm preserved (and slightly increased) lean mass.^[10]

Resistance training during the deficit. The Murphy and Koehler (2022) meta-analysis showed that an energy deficit of ~500 kcal/day blunts but does not eliminate lean-mass gains from resistance training; strength gains persist regardless of deficit.^[11] The 2025 Frontiers review reported that the majority of resistance-training participants in pooled weight-loss trials maintained or gained lean mass.^[13]

About the magnitude of the drop.

A direct GLP-1 + protein + resistance-training intervention trial reporting the resulting lean-mass-loss share has not yet appeared in the published literature as of June 2026. The directional claim above is based on adjacent (non-GLP-1) intervention trials and observational data. When such a trial publishes, the directional language will be replaced with the cited figure.

What Muscle Guard tracks for you.

Protein adherence — log meals via AI plate scan; see daily intake vs your weight-adjusted target.

Resistance training frequency and volume — log sessions, sets, weight per exercise.

Body composition trend — waist, hip, body-fat estimate, visceral-fat rating.

Muscle Guard Score — composite metric capturing fat-loss-without-muscle-loss.

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BRANCH 1 · WHAT GLP-1s DO TO YOUR BODY

GLP-1 side effects, week by week.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 9-minute read

Approximately 40-70 percent of users on semaglutide or tirzepatide experience gastrointestinal side effects at some point, most commonly during titration steps.^{[19][18]} The pattern matters more than any single symptom. This article walks through the typical timeline and the standard titration schedules.

Weeks 1-4 — starting dose.

Standard semaglutide starting dose is 0.25 mg weekly. Tirzepatide is 2.5 mg weekly. Most users report mild to moderate nausea, fatigue, appetite suppression by week 2, and constipation.^[18]

Weeks 5-12 — first titration steps.

Standard semaglutide titration: 0.25 mg → 0.5 mg at week 5 → 1.0 mg at week 9. Tirzepatide titrates over 16-20 weeks.^{[2][4]} Nausea returns at each step-up; usually milder than week 1 but real.

Weeks 13-24 — approaching peak dose.

Semaglutide approaches 1.7 mg then 2.4 mg in this window; tirzepatide reaches 12.5-15 mg. Most users settle at peak dose somewhere here.^[23]

GI symptoms generally easier than during escalation

Plateaus possible — weight loss is rarely linear

'Food blankness' — nothing sounds appealing

Muscle loss becomes visible if not actively prevented

When to call your healthcare provider.

Severe abdominal pain, especially if persistent — possible pancreatitis.

Vomiting that prevents keeping liquids down.

Signs of dehydration — dark urine, dizziness on standing.

Persistent or worsening fatigue.

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BRANCH 1 · WHAT GLP-1s DO TO YOUR BODY

Mounjaro vs Ozempic vs Wegovy: a 2026 evidence comparison.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 10-minute read

In head-to-head trials, tirzepatide produces larger average weight loss than semaglutide at maximal doses: ~21 percent vs ~15-17 percent over 68-72 weeks.^{[4][2][7]} Side-effect profiles are broadly similar; cost and regional access differ substantially.

At a glance.

Ozempic and Wegovy are both semaglutide. **Mounjaro and Zepbound** are both tirzepatide. Ozempic and Mounjaro are registered for type 2 diabetes; Wegovy and Zepbound for chronic weight management.

Efficacy.

Wegovy 2.4 mg vs placebo: 14.9 percent body weight loss at week 68 (STEP-1).^[2]

Tirzepatide 15 mg vs placebo: 20.9 percent body weight loss at week 72 (SURMOUNT-1).^[4]

Head-to-head in T2D (SURPASS-2): tirzepatide arms produced larger loss than semaglutide.^[7]

Body composition.

DEXA substudies show meaningful lean-mass loss in absence of intervention.^{[5][4]} The Neeland 2024 review notes that the absolute lean-mass loss is larger with tirzepatide because total weight loss is larger.^[1]

Cost and access (2026, regional).

South Africa. Ozempic registered for T2D, R2,700-R3,100/month. Wegovy not yet registered. Compounded common at R1,000-R2,500/month.

United States. Wegovy and Zepbound FDA-approved for weight management. Insurance coverage variable.

European Union. Wegovy available in most member states; access varies by country.

Muscle preservation.

None of the three drugs preserve muscle by themselves. The lifestyle intervention — protein at 1.2-1.6 g/kg/day and resistance training 2-4 times per week — is the variable that substantially reduces the lean-mass-loss share.

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BRANCH 1 · WHAT GLP-1s DO TO YOUR BODY

Compounded semaglutide and tirzepatide: what the evidence says.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 10-minute read

Compounded semaglutide is real semaglutide — formulated by a registered compounding pharmacy. The FDA has flagged significant safety concerns including dosing errors, sterility, and adverse events as of mid-2024 through 2025.^{[14][15]} In South Africa, where branded products are often unaffordable, compounded preparations remain mainstream.

What the FDA has flagged.

Dosing errors. July 2024: patient inexperience with measuring and self-administering compound semaglutide led to overdoses of 5 to 20 times the intended dose.^[14]

Adverse events. As of December 31, 2024, the FDA's Adverse Event Reporting System recorded more than 900 cases including 17 deaths.^[15]

Sterility. The FDA has expressed concerns about the sterility of products from some compounding facilities.

Quality variation. Compounded semaglutide has no FDA approval; benefit and risk profiles are not equivalent to the branded product.

The South African context.

Compounded semaglutide and tirzepatide from registered SAPC compounding pharmacies remain widely used. Compounded preparations cost R1,000-R2,500/month vs R2,700-R3,100 for branded Ozempic.^[22] Quality varies by pharmacy.

How to assess a compounding pharmacy.

SAPC (SA) / state pharmacy board (US) registration.

API source disclosure and certificate of analysis available.

Formulation transparency — semaglutide salt form disclosed.

Refrigerated transport and clear storage instructions.

Pharmacist accessibility — they answer questions.

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BRANCH 1 · WHAT GLP-1s DO TO YOUR BODY

Rybelsus and oral semaglutide: how it works, dosing rules.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 8-minute read

Rybelsus is oral semaglutide co-formulated with absorption enhancer SNAC.^[17] Bioavailability is approximately 0.4-1 percent, which is why the dose is much higher than the injectable equivalent.^[16] The 30-minute pre-meal fasting rule is non-negotiable.

What Rybelsus actually is.

Oral semaglutide co-formulated with SNAC (sodium N-[8-(2-hydroxybenzoyl)amino] caprylate), an absorption enhancer that facilitates semaglutide crossing the gastric epithelium.^[17]

Dosing schedule.

Weeks 1-4: 3 mg daily.

Weeks 5-8: 7 mg daily.

Week 9+: 14 mg daily.

The 30-minute fasting window.

Take on an empty stomach with no more than 120 mL of water. Then nothing — no food, drink, or other oral medications — for 30 minutes. Coffee, fruit juice, oat milk, or multivitamins within the window substantially reduce the dose reaching the bloodstream.^[17]

Side effects.

Mirrors injected semaglutide — nausea, fatigue, constipation. Oral produces less peak-trough fluctuation because the daily dose smooths the blood-concentration curve.^[16]

Muscle preservation.

Daily dosing doesn't change the muscle preservation requirement. Protein 1.2-1.6 g/kg/day; resistance training 2-4 times per week.

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

The protein target: 1.2 to 1.6 g/kg per day, in plain English.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 9-minute read

For adults on a GLP-1, the evidence supports a daily protein target of 1.2-1.6 grams per kilogram of body weight — meaningfully above the 0.8 g/kg/day general RDA.^{[8][21]} The Longland et al. (2016) trial showed that the upper end of this range, combined with resistance training, preserves and even slightly increases lean mass during a substantial energy deficit.^[10]

Why the target is higher on a GLP-1.

The 0.8 g/kg/day RDA was set to prevent deficiency in sedentary, weight-stable adults. People on a GLP-1 are neither sedentary nor weight-stable.^{[21][9]}

Translating the range.

60 kg adult — 72 to 96 g protein per day.
75 kg adult — 90 to 120 g protein per day.
90 kg adult — 108 to 144 g protein per day.

Per-meal target.

Spread protein across three or four meals containing 25-40 g each. Each meal fully stimulates muscle protein synthesis for several hours. Skipping breakfast cuts opportunities by 25-33 percent.

Practical sources.

2 eggs — 12 g · 100 g chicken breast — 31 g · 100 g salmon — 25 g
200 g Greek yoghurt — 20 g · 1 scoop whey protein — 24 g · 100 g cottage cheese — 11 g
100 g lentils (cooked) — 9 g · 100 g firm tofu — 13 g · 1 tin tuna (drained) — 25 g

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

Resistance training on a GLP-1: the minimum effective dose.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 8-minute read

Two to four resistance sessions per week, at RPE 7 to 9, with progressive overload. That's the minimum effective dose the published intervention arms used to substantially reduce lean-mass-loss share below the 25-40 percent un-intervened baseline.^{[11][13]} The protocol is straightforward — what matters is the consistency.

Why resistance training, specifically.

The body has no reason to preserve muscle in a sustained calorie deficit unless something tells it to. Aerobic exercise does not provide that signal at meaningful intensity. Resistance training does.

The minimum effective dose.

Frequency: 2-4 sessions per week.

Volume: 10-20 hard sets per muscle group per week.

Intensity: RPE 7-9.

Progression: add weight, reps, or sets every one to two weeks.

The Schoenfeld et al. (2019) meta-analysis on training frequency found that 2x/week per muscle group is sufficient for hypertrophy when weekly volume is matched.^[12]

What 'a session' actually looks like.

45-60 minutes, all major muscle groups, compound movements as the spine: squat or leg press; deadlift; bench press or push-up progression; row or pull-up progression; overhead press.

Cardio is welcome, but secondary.

Cardiovascular work is useful for cardiovascular health, mood, and additional energy expenditure. It does not replace resistance training for muscle preservation.

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

Body recomposition on a GLP-1: can you build muscle while losing fat?

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 8-minute read

True recomposition — gaining lean mass while losing fat — is harder in a sustained calorie deficit than in normal training conditions, but not impossible.^{[10][11]} For most users on a GLP-1, the realistic goal is closer to 'minimise muscle loss while losing fat' with periods of genuine recomposition during dose plateaus.^[1]

What recomposition means in this context.

On a GLP-1 you are nearly always in a calorie deficit. Longland et al. (2016) demonstrated lean-mass gain with 2.4 g/kg/day protein and concurrent training during a 4-week 40 percent deficit in young men.^[10] For the typical GLP-1 user (longer time horizon, less extreme protein intake), the data supports 'minimise muscle loss' more than 'gain muscle' — and the share of weight lost from lean mass drops substantially below the un-intervened baseline.

Tracking cadence.

Weight: weekly.

Waist circumference: weekly, same time of day.

Body composition (DEXA or BIA): every 12 weeks while losing.

Progress photos: monthly, same angles, same lighting.

What 'good' looks like at 12 weeks.

Users following the protein-and-training plan typically show: weight down 8-12 percent; waist circumference down 5-10 cm; body-fat estimate down 4-6 percentage points; strength on baseline lifts holding or improving.

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

Muscle drain and 'Ozempic face': what's actually happening.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 7-minute read

'Muscle drain' and 'Ozempic face' are user-coined terms for the same root phenomenon — losing lean mass alongside the fat. The mechanism is a sustained calorie and protein deficit without a resistance-training stimulus.^{[1][9]} Both are largely preventable with the protein-and-training plan.

What 'muscle drain' actually describes.

Without intervention, between roughly a quarter and 45 percent of weight lost on a GLP-1 can be lean tissue rather than fat.^{[1][5][4]} Some is skeletal muscle, some is connective tissue, glycogen, and water.

Why it shows up most in users who lose weight fastest.

Faster weight loss requires a larger calorie deficit, which drives the body harder toward mobilising both fat and muscle stores.

The face specifically.

The face has fat pads (buccal, malar, periorbital) that contribute to the soft, rounded look associated with 'healthy' weight. Rapid loss in these depots is part of the 'Ozempic face' appearance.

The prevention plan.

Protein: 1.2 to 1.6 g/kg/day.

Resistance training: 2-4 sessions per week.

Don't push the deficit: 0.5-1.0 percent body weight loss per week is sustainable.

Track waist circumference: A flat-scale week with a tightening waist is recomposition.

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

Maintenance after a GLP-1: what the research says.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 8-minute read

In the STEP-1 trial extension, participants regained roughly two-thirds of the lost weight within 12 months of semaglutide discontinuation.^[3] Net weight loss from baseline shrank from 17.3 percent on-treatment to 5.6 percent one year off. Users who continued the protein-and-training plan during the medication phase regain less.

Why people stop.

Reasons vary by region. In the US: insurance coverage cycles, plan changes. In SA: cost, supply. In the EU: stigma, side-effect intolerance, pregnancy.

Regain patterns from the published data.

Full discontinuation, no plan: 65-80 percent of lost weight regained over 12 months.^[3]

Discontinuation with protein + training continued: regain is smaller and more body-composition-neutral.

Taper to low maintenance dose: regain is smallest, with prescriber agreement.

Taper or hard stop.

Both are legitimate. Hard stop is most users' default. Taper (step down dose over 8-16 weeks) gives a smoother appetite transition.

Muscle preservation in maintenance is non-negotiable.

The single biggest determinant of how the post-GLP-1 year goes is whether you continued resistance training during the medication phase.

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

The Muscle Guard Score, explained.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 7-minute read

The Muscle Guard Score is one number, 0-100, that captures whether the fat-loss-without-muscle-loss plan is working. It combines four inputs — protein adherence, resistance training frequency, weight trend, and body composition — weighted by their relative contribution to muscle preservation evidence.

Why one number, not five.

The question 'am I doing this properly?' is composite. It requires weighting protein against training against scale movement against body composition simultaneously. Humans aren't good at composite mental scoring; we anchor on the metric that moved most this week and miss the one that quietly drifted.

The four inputs and their weights.

Protein adherence — 35 percent. Rolling 7-day average of intake vs target.

Resistance training frequency — 25 percent. Rolling 14-day count against target.

Weight trend — 20 percent. 14-day rolling rate.

Body composition — 20 percent. 30-day direction of travel.

Score bands.

0-49: needs attention.

50-69: okay.

70-84: on track.

85+: excellent.

Diagnostic, not graded — the goal is consistency in the 70+ range, not chasing 100.

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BRANCH 3 · THE RESEARCH, UPDATED

GLP-1s in South Africa, 2026: cost, access, the compounded landscape.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 8-minute read

Branded Ozempic costs R2,700-R3,100 per month in South Africa as of 2026 and is registered with SAHPRA for type 2 diabetes only.^[26] Medical aid coverage for weight-loss use is rare. Compounded semaglutide and tirzepatide from registered SAPC pharmacies fill the gap at R1,000-R2,500 per month.^[22]

What Ozempic actually costs.

Ozempic 0.25 mg (starter): R1,800-R2,200/month.

Ozempic 0.5 mg: R2,300-R2,700/month.

Ozempic 1.0 mg: R2,700-R3,100/month.

Compounded semaglutide: R1,000-R2,500/month.

Sources: Dis-Chem and Clicks public price lists Q1 2026; Spotlight SA reporting.^[22]

Medical aid coverage.

For diabetes (registered indication): covered with a chronic-condition motivation.

For weight loss alone: almost universally not covered.

For weight loss with comorbidity: occasionally approved with BMI > 35 plus documented comorbidities.

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BRANCH 3 · THE RESEARCH, UPDATED

Getting the most from your GLP-1 doctor visit.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 6-minute read

Most GLP-1 consults are 15-20 minutes. The productive ones start with a one-page summary your prescriber reads in 60 seconds, three written questions, and a current medication list. They end with three clear actions you've logged before you leave the parking lot.

The five-day prep window.

Day -5: Run a draft Doctor PDF.

Day -4 to -2: Fix the gaps.

Day -1: Write three questions.

Morning of: Generate the final PDF.

Three questions to ask every consult.

Looking at the trend, is the dose right for the next 4-6 weeks?

Which side effect should I track most carefully?

What would make you change the plan?

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BRANCH 3 · THE RESEARCH, UPDATED

Privacy and GLP-1 apps: what your data does on the leading trackers.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 7-minute read

GLP-1 trackers process at least three categories of personal data: identifying information; behavioural information (weight, food, exercise, photos); and inferred information (your patterns, risk flags, recommendations). Health data is 'special category' under Article 9 GDPR — requiring explicit consent or another specific legal basis.

Your rights, in plain language.

Access — get a copy of every piece of data the app holds about you.

Rectification — correct anything inaccurate or incomplete.

Erase — delete your data, the account, the lot, usually under 30 days.

Portability — export your data in a machine-readable format.

How to read a privacy policy in 10 minutes.

Search for 'third party' / 'advertising' / 'international transfer' / 'retention' / 'legal basis'.

How Muscle Guard handles privacy.

EU-region data hosting for EU users; equivalent posture for ZA and US.

Explicit consent for health data processing at sign-up.

No third-party advertising. No data sale.

Deletion in one tap. All personal data removed within 7 days.

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Citations.

All sources used across this library. Click any DOI/URL to open the original paper.

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Methods note.

Sources are peer-reviewed unless otherwise labelled. Regulatory body publications (FDA, SAHPRA) and journalism sources (Spotlight SA) are used for market and access claims, labelled inline. The 25-40 percent un-intervened lean-mass-loss share is directly cited. The 'drops substantially' phrasing for post-intervention is supported by adjacent (non-GLP-1) intervention trials; a GLP-1-specific intervention trial reporting an exact post-intervention figure is pending publication and will be cited here when it lands.

Errors.

Email research@muscleguardglp.com. We update this PDF when corrections are warranted and bump the version date in the footer.